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Beyond Patches: Global-aware Autoregressive Model for Multimodal Few-Shot Font Generation

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Abstract

Manual font design is an intricate process that transforms a stylistic visual concept into a coherent glyph set. This challenge persists in automated Few-shot Font Generation (FFG), where models struggle to preserve both structural integrity and stylistic fidelity from limited references. While autoregressive (AR) models have demonstrated impressive generative capabilities, their application to FFG is constrained by conventional patch-level tokenization, which neglects global dependencies crucial for coherent font synthesis. Moreover, existing FFG methods remain within the image-to-image paradigm, relying solely on visual references and overlooking the role of language in conveying stylistic intent during font design. To address these limitations, we propose GAR-Font, a novel AR framework for multimodal few-shot font generation. GAR-Font introduces a global-aware tokenizer that effectively captures both local structures and global stylistic patterns, a multimodal style encoder offering flexible style control through a lightweight language-style adapter without requiring intensive multimodal pretraining, and a post-refinement pipeline that further enhances structural fidelity and style coherence. Extensive experiments show that GAR-Font outperforms existing FFG methods, excelling in maintaining global style faithfulness and achieving higher-quality results with textual stylistic guidance. Project Page: https://xtryer-s.github.io/projects_pages/GAR_Font

1. Introduction

High-quality fonts are central to visual communication, yet their manual creation is costly: practitioners must translate their design concepts into a full, consistent glyph set. This challenge is compounded for logographic systems like Chinese and Japanese, which contain tens of thousands of characters and complex stroke geometries.



Figure 1. GAR-Font results under visual and multimodal few-shot settings. The generated poem reflects the key contributions of our model: global-aware tokenization for style fidelity, multimodal style encoding for text-image control, reduced reference requirements, and an autoregressive design that enables controllable high-quality font synthesis.

Few-shot Font Generation (FFG) seeks to automate this process by generating an entire font library from only a handful of reference examples. However, it presents several technical difficulties. First, it demands precise structural fidelity, where strokes, radicals, and local geometry must be correct for every glyph. Second, it requires capturing a holistic and faithful style representation, generalizing from a handful of examples across diverse characters.

A core obstacle for FFG is the lack of an effective visual global representation. GAN-based methods [29, 46] often exhibit noticeable style discrepancies from the reference fonts, and struggle with stroke-level accuracy. Diffusion models [21, 33, 71] improve structural correctness and local fidelity but do not guarantee a coherent global style. Existing sequence approaches such as VQ-Font [72] and IF-Font [13] rely on local patch or block tokenization, which fragments global cues and leads to noticeable discrepancies from the reference styles. Recent autoregressive (AR) image generation methods [25, 75] further reveal the superi-

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ority of globally contextualized 1D tokens over 2D patched tokenization, highlighting a promising direction for modeling global stylistic patterns in font synthesis.

Meanwhile, existing FFG methods remain single-modal, using only visual modality for style control. In contrast, linguistic descriptions convey global conceptual design intents beyond visual appearance, providing complementary representations that help overcome the limitations of vision-only models when style must be inferred from a few examples.

These insights motivate GAR-Font, an expressive and controllable FFG framework that integrates visual and linguistic information. Leveraging the AR modeling capacity, GAR-Font introduces three major technical contributions:

- We propose **a global-aware tokenizer (G-Tok)** that fuses local features with global perception, capturing both fine-grained stroke details and font-level style patterns.
- We design **an AR generator with multimodal style encoder**, first pretrained on visual inputs, and then augmented with a lightweight language-style adapter.
- We introduce **a post-refinement pipeline** that improves structural fidelity and stylistic coherence, producing high-quality glyphs from limited references.

To validate the effectiveness of GAR-Font, we conduct extensive experiments across multiple settings. In the standard vision-only scenario, it surpasses other competitive FFG methods, demonstrating superior structural and stylistic fidelity. In the multimodal setting, the augmented multimodal style encoder further improves generation quality. Using 4 reference images plus one textual description, GAR-Font matches the quantitative performance of 8 images while providing flexible control. Ablation studies further confirm that G-Tok produces stable and coherent representations that preserve holistic structure and reference style fidelity—key factors for high-quality font synthesis. GAR-Font moves beyond patch-level modeling by learning global-aware representations and enabling text-driven multimodal FFG, establishing a new AR-based solution for automatic and controllable font generation.

2. Related Work

2.1. Autoregressive Image Generation

Autoregressive (AR) models have shown powerful generation ability across modalities [1, 32, 38, 42], and their extension to visual generation [14, 48, 52, 53, 60, 61, 74] has demonstrated promising potential. Typically, they follow a two-stage paradigm: a tokenizer encodes images into discrete representations (e.g., VQ-VAEs [49, 58], RQ-VAE [31], VQ-GAN [17, 73]), and a transformer decoder [10, 25, 44, 57, 76] that sequentially predicts these tokens for image reconstruction.

Most AR models use 2D patch-wise tokenization, capturing local details but lacking holistic awareness [5, 12, 24, 64]. To improve global coherence, recent works explore complementary strategies: global-context modeling [80] introduces holistic queries for structural reasoning; frequency-domain methods [25, 26] encode coarse-to-fine spectral context; 1D tokenizers [2, 65, 75] enhance efficiency but lose adaptability; semantic tokenization [35, 77] leverages language priors for meaningful visual codes. Structured visual like glyphs, however, demands tokenizers that unify local stroke detail with global aesthetics. Our GAR-Font addresses this challenge through a **global-aware tokenizer** that integrates CNN-based locality with Transformer-based global reasoning, effectively unifying fine-grained stroke detail and overall stylistic coherence.

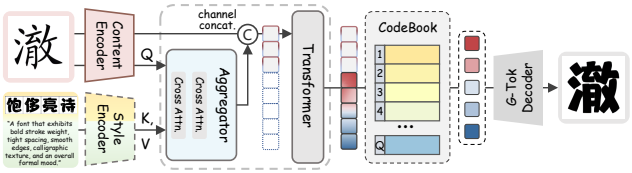


Figure 2. The overall architecture of GAR-Font. It comprises a global-aware tokenizer (G-Tok), and an AR generator, equipped with a multimodal style encoder.

2.2. Few-shot Font Generation

Few-shot Font Generation (FFG) can be broadly divided into vector-based and image-based approaches. Vector FFG methods have progressed in sequential modeling and efficient representations [41, 55, 56, 62, 63, 68]. Yet complex ideographic generation and unseen glyph generalization remain challenging [36, 37]. Image-based methods, in contrast, learn 2D pixel priors for robust adaptation, evolving from early image-to-image translation [4, 9, 11, 66] to more recent VQ and diffusion formulations [33, 34, 71, 72]. These models typically employ content–style disentanglement to separately capture structure and style [8, 19, 21, 22, 39, 45, 47, 70, 78, 79], enhanced by mechanisms like contrastive learning for global coherence [27] and cross-attention for local transfer [54].

The challenge is most acute for glyph-rich, ideographic languages such as Chinese and Korean, where thousands of characters demand fine structure and faithful global style [18, 27, 28, 33]. In this regime, **image-based AR models that operate FFG on globally contextualized visual tokens** show particular promise: by modeling spatial dependencies directly in the 2D domain, they better capture the intricate geometry and visual regularities of glyphs.

2.3. Multimodal Alignment for Style Control

Multimodal alignment for style control unifies visual and textual modalities for coherent, fine-grained style manipulation. Foundational models [6, 14, 15, 20, 61, 67] achieve this through costly multimodal pretraining. For fonts, how-

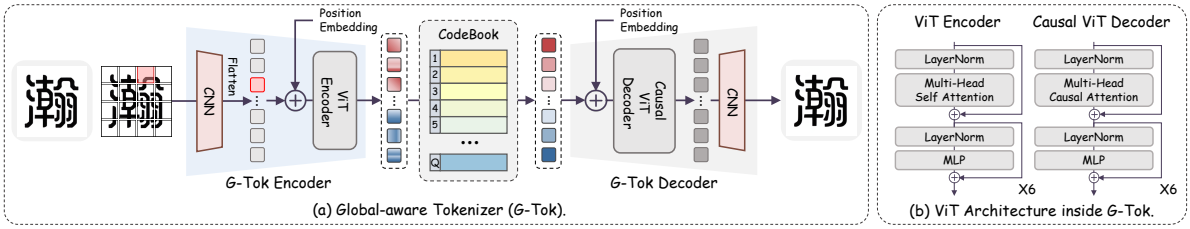


Figure 3. (a) Overview of the G-Tok architecture, which adopts a hybrid CNN–ViT design. (b) Details of the global ViT encoder and causal ViT decoder. With self- and causal-attention, G-Tok captures global dependencies, enabling coherent and high-quality font synthesis.

ever, only textual inputs describing style are incorporated, and the independent encoding of content and style in FFG frameworks [46, 54, 72] naturally forms a structured visual style space that serves as a strong prior for multimodal integration. Recent works pursue efficient alignment via parameter-efficient modules [7, 23, 30, 51] or resampler-based strategies aligning vision and language spaces [1, 40, 43, 69]. Motivated by these works, GAR-Font extends the style encoder in FFG models to a **multimodal style encoder** that consists of a visual style encoder and a lightweight pluggable **language-style adapter** that aligns textual descriptions with visual style embeddings, enabling fine-grained, flexible style control without the need for large-scale multimodal pretraining.

3. Method

The architecture of GAR-Font is illustrated in Figure 2. Our framework comprises three core components:

A global-aware tokenizer (G-Tok) discretizes glyphs into tokens, capturing intricate structure and global visual style.

An AR generator with multimodal style encoder first learns from visual inputs to establish a stable style space, and then employs a lightweight language-style adapter for multimodal style control.

A post-refinement enhances structural fidelity and style coherence for high-quality few-shot font generation.

Given a content glyph, a few style references, and optional text, GAR-Font encodes and aggregates content and style, autoregressively generates codebook representations and decodes them softly into high-quality glyphs, which are subsequently improved by a post-refining stage. The details of each module adopted in the proposed GAR-Font will be presented in the following subsections.

3.1. Global-aware Tokenizer

The Global-aware Tokenizer (G-Tok), shown in Figure 3, encodes glyphs into tokens using a hybrid CNN–ViT encoder, a vector quantizer, and a causal hybrid decoder. By fusing local convolutional features with Transformer-based global context, it captures fine-grained structure and overall style, enabling coherent, high-quality font synthesis.

Hybrid encoding. Given a glyph image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, a CNN encoder E_{CNN} extracts local stroke features, which are flattened and aggregated through a ViT encoder E_{ViT}:

$$\mathbf{T} = E_{\text{ViT}}\big(\text{Proj}(E_{\text{CNN}}(\mathbf{I})) + \mathbf{P}_{\text{2D}}\big) \in \mathbb{R}^{N\times d}, \quad (1)$$

where $\mathbf{P}_{\text{2D}}$ refers to the 2D sinusoidal position embeddings [16], N and d denote token count and embedding dimension. The CNN backbone preserves spatial locality, capturing fine stroke geometry, while the ViT aggregates tokens globally for style fidelity. This hybrid design captures both structure fidelity and long-range style dependencies.

Vector Quantization. Following standard VQ-VAE [58] and VQ-GAN [17], we discretize latent tokens by mapping them to the nearest entries in a learnable table. We apply the commitment and embedding regularization with an entropy term to stabilize training and preserve diversity.

Causal decoding and optimization. A causal ViT–CNN decoder reconstructs glyph $\hat{\mathbf{I}}$ from quantized tokens, modeling sequential dependencies while convolutional layers refine local details. Figure 3(b) illustrates this structure.

The tokenizer is optimized end-to-end with weighted reconstruction, perceptual, and vector quantization losses:

$$\mathcal{L}_{\text{tok}} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{per}}\mathcal{L}_{\text{per}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}}, \quad (2)$$

where $\mathcal{L}_{\text{rec}} = \|\mathbf{I} - \hat{\mathbf{I}}\|_1$ denotes the L1 reconstruction loss, $\mathcal{L}_{\text{per}} = \|\Phi(\mathbf{I}) - \Phi(\hat{\mathbf{I}})\|_2^2$ represents the perceptual loss, and $\mathcal{L}_{\text{vq}}$ is the standard vector quantization loss.

3.2. AR Generator with Multimodal Style Encoder

Using G-Tok's global representations, the AR generator performs conditional sequential prediction. To leverage prior FFG insights (Content-style Aggregator) while avoiding the cost of joint image–text training, GAR-Font adopts a decoupled learning paradigm. The generator, aggregator, and style encoder are first trained on visual inputs to learn a stable representation. Based on this foundation, the visual style encoder is extended into a multimodal one through a lightweight, plug-in language adapter that aligns textual cues with learned visual styles, enabling flexible text-guided generation without disturbing the visual prior.

3.2.1. Pretraining on Visual Modality

Visualized in Figure 4(a), to build a stable style–content representation, the AR generator is first trained on visual conditions, following the successful practices of prior FFG methods. This ensures robust modeling of both content structure and stylistic variations across glyphs, providing a solid foundation for subsequent multimodal adaptation.

Encoders and Content–style Aggregator. As shown in Figure 4(a), we adopt both CNN architectures on the content encoder and the visual style encoder. The Content-style Aggregator follows the previous design [46, 54]. Given a content glyph and N_s style references, the encoders respectively extract content features $\mathbf{F}_c$ and style features$\{\mathbf{F}_{vis_j}\}_{j=1}^{N_s}$. These features are then fused where content queries attend to fine-grained style cues:

$$\tilde{\mathbf{T}}_{\text{vis}} = \text{Aggregator}\big(\mathbf{F}_c, \{\mathbf{F}_{\text{vis}_j}\}_{j=1}^{N_s}\big). \tag{3}$$

The aggregated visual representation $\tilde{\mathbf{T}}_{\text{vis}}$ is concatenated with $\mathbf{F}_c$ to form the conditioning input $\mathbf{T}$.

Autoregressive modeling and soft decoding. Conditioned on $\mathbf{T}$, the Transformer generator autoregressively predicts the next token, producing logits $\mathbf{L}$ over the G-Tok codebook $\mathcal{C}$. Instead of discrete hard decoding, we apply a soft projection that maps $\mathbf{L}$ onto the codebook $\tilde{\mathbf{Z}} = \text{Softmax}(\mathbf{L}) \cdot \mathcal{C}$. This continuous mapping not only preserves gradient flow during training, enabling pixel-level supervision, but can also improves stroke continuity and glyph fidelity by leveraging the full expressive capacity of the codebook, yielding smoother and more accurate glyphs compared with discrete hard decoding.

Training objectives. During this visual-pretraining stage, the aggregator, generator, and style encoder are jointly optimized with a combined token and pixel-level loss:

$$\mathcal{L}_{\text{AR}} = \lambda_{\text{CE}}\mathcal{L}_{\text{CE}} + \lambda_{\text{pixel}}\mathcal{L}_{\text{pixel}}, \tag{4}$$

where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss over the target token indices $\mathbf{s}$, and $\mathcal{L}_{\text{pixel}}$ measures the L1 reconstruction error between the generated glyph $\hat{\mathbf{I}}_f$ and ground truth $\mathbf{I}_f$.

3.2.2. Multimodal Style Encoder Adaptation

Visual pretraining forms a stable style–content space but lacks high-level conceptual control. To address this, we introduce a lightweight, plug-in language adapter that aligns textual design cues with visual style representations, as shown in Figure 4(b). Through this adapter, the style encoder is extended into a multimodal form, enabling flexible text-guided modulation without disrupting the visual prior.

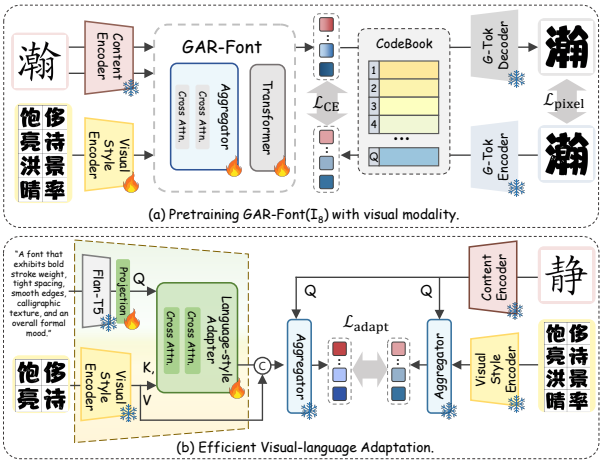


Figure 4. GAR-Font adopts a two-stage training: (a) Visual Pretraining builds a stable content–style space via token and pixel losses; (b) Vision–language Adaptation aligns text embeddings with style features with a lightweight adapter for text-guided generation while preserving visual priors.

Vision-language Adapter. The adapter bridges textual descriptions and visual style features through iterative cross-attention. Specifically, a subset of visual style features $\{\mathbf{F}_{vis_j}\}_{j=1}^{k}$ is first extracted from $k < N_s$ reference glyphs by the visual style encoder, while a textual font description embedding is derived from a pretrained Flan-T5 encoder. The text embedding is projected into the visual feature space and iteratively refined by attending to style features. The aligned textual–visual token is then spatially expanded into $\mathbf{F}_t$ and concatenated with the visual features:

$$\tilde{\mathbf{F}}_{mm} = [\mathbf{F}_{vis_1}, \ldots, \mathbf{F}_{vis_k}, \mathbf{F}_t]. \tag{5}$$

Training objectives. The multimodal style encoder may produce features $\tilde{\mathbf{F}}_{mm}$ of different token lengths than the visual-only encoder due to varying numbers of visual references. Thus, we supervise training on their aggregated objectives. Specifically, $\tilde{\mathbf{T}}_{\text{mm}}$ and $\tilde{\mathbf{T}}_{\text{vis}}$ are obtained by aggregating k visual plus one textual reference, and all N_s visual references, respectively. The adapter is trained by minimizing the ℓ_2 distance between these representations:

$$\mathcal{L}_{\text{adapt}} = \|\tilde{\mathbf{T}}_{\text{mm}} - \tilde{\mathbf{T}}_{\text{vis}}\|_2^2. \tag{6}$$

This objective encourages the multimodal encoder to internalize stylistic intent consistent with the full-visual setting, facilitating effective text–style substitution and reducing reliance on visual inputs.

3.3. Post-refinement

GAR-Font adopts a two-stage post-refinement to improve few-shot style generalization and structural accuracy: novel font adaptation (NFA) and structural enhancement (SE).

Novel font adaptation (NFA) for few-shot generalization. The pretrained generator learns general font patterns but exhibits minor inconsistencies on unseen styles. NFA mitigates this by performing a lightweight adaptation using a few reference glyphs, updating the LoRA layers of the Transformer generator with a mixed token–pixel loss:

$$\mathcal{L}_{\text{NFA}} = \lambda_{\text{CE}}\mathcal{L}_{\text{CE}} + \lambda_{\text{pixel}}\mathcal{L}_{\text{pixel}}, \tag{7}$$

where $\mathcal{L}_{\text{CE}}$ is the token cross-entropy and $\mathcal{L}_{\text{pixel}} = \|\mathbf{I} - \hat{\mathbf{I}}\|_1$ encourages pixel-level accuracy. This yields stable few-shot adaptation and better preservation of unseen styles.

Structural enhancement (SE) for precise glyph reconstruction. While NFA enhances style fidelity, minor structural distortions may remain. To enhance structural clarity and readability, SE further refines glyphs using a group-relative optimization based on GRPO [50]. The generator is treated as a policy π_θ that outputs token sequences $\mathbf{s}$; each decoded glyph receives a composite reward:

$$r = \lambda_{\text{ocr}}\, r_{\text{ocr}} + \lambda_{\text{style}}\, r_{\text{style}}, \tag{8}$$

where r_{ocr} is obtained from a pretrained OCR model as

$$r_{\text{ocr}} = \begin{cases} p_{\text{ocr}}, & \text{if } \hat{y} = y, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Here p_{ocr} is the recognition confidence, and r_{style} measures the style consistency via a pretrained discriminator.

Rewards are normalized within each sampled group to compute advantages $A^{(k)} = \frac{r^{(k)} - \mu(r)}{\sigma(r)}$. SE updates only the LoRA layers by maximizing advantage-weighted likelihood with KL regularization to a frozen reference policy:

$$\mathcal{L}_{\text{SE}} = -\mathbb{E}_{\mathbf{s}\sim\pi_\theta}\left[A(\mathbf{s}) \log \pi_\theta(\mathbf{s})\right] + \beta\, \text{KL}(\pi_\theta \| \pi_{\text{ref}}), \tag{10}$$

where $A(\mathbf{s})$ denotes the group-normalized advantage and π_{ref} is the frozen reference policy used for stability.

4. Experiments

4.1. Datasets and Evaluation Metrics

We evaluate GAR-Font and prior FFG methods on two datasets: a small-scale set containing 440 font styles (*S*) and a large-scale set with 3,040 styles (*L*). The small-scale dataset is a strict subset of the large-scale one, allowing fair cross-scale comparison. In each dataset, we randomly select 40 as unseen test fonts, while the remaining 400 (for *S*) or 3,000 (for *L*) are used for training.

Both datasets are constructed based on the official GB2312 character set, including 6,763 Chinese characters. Among them, 6,251 characters are randomly chosen for training, and the remaining 512 are held out as unseen characters. During evaluation, we consider two settings:
(1) *Unseen Fonts Seen Characters (UFSC)*, where 512 seen characters are rendered with the 40 unseen fonts; and
(2) *Unseen Fonts Unseen Characters (UFUC)*, where 512 unseen characters are rendered with the same unseen fonts.

We use RMSE↓, SSIM↑, LPIPS↓, and FID↓ to measure pixel- and perception-level similarity between generated and ground-truth glyphs. Following [33, 46], we train a content classifier over 6,763 characters (99.71% accuracy) and a style classifier over 3,040 fonts (92.72% accuracy) to compute content accuracy (Acc(C)↑) and style accuracy (Acc(S)↑). All glyphs are resized to 64×64 for evaluation.

4.2. Implementation Details

The G-Tok tokenizer discretizes each glyph into 64 tokens using a 2,048-entry, dimension-8 codebook, trained for 200k iterations (batch = 16, lr = 1×10^{-4}). The vision-only AR generator uses the AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.95$), conditioned on one Kaiti content. GAR-Font(I_8) is trained with $N_s = 8$ style glyphs for 600k/1M iterations on the small/large sets (batch = 32, lr = 1×10^{-4}).

For multimodal style encoder adaptation, the Language–style Adapter is trained for 40k iterations (batch size = 128, lr = 1×10^{-4}) with the visual-only features ($N_s = 8$) as ground truth. By adjusting multimodal encoder's visual style reference numbers $k = 2/4$, we build multimodal variants: GAR-Font(M_2) and GAR-Font(M_4).

In post-refinement, NFA is conducted on 8 target font glyphs (denoted as NFA-8) for 10 epochs (lr = 2×10^{-5}). SE is based on GRPO, where each group generates 4 samples and each character is trained with 8 glyphs from different fonts (epochs = 10, batch size = 32, learning rate = 5e−6).

4.3. Comparison on Few-shot Font Generation

We conduct an experiment for **vision-only FFG**, where all models generate glyphs from only reference images. GAR-Font(I_8) is compared with seven open-source methods: GAN/VAE-based LF-Font [46], VQ-Font [72], DG-Font [70], CF-Font [59]; diffusion-based Diff-Font [21] and Font-Diffuser [71]; and the AR-based IF-Font [13]. All methods are trained and evaluated on the *S* and *L* datasets using their official resolutions and hyperparameters. GAR-Font(I_8) is evaluated both before and after post-refinement.

Table 1 shows that GAR-Font(I_8) consistently outperforms existing vision-only FFG approaches. Even at the pretrained stage, it achieves competitive RMSE↓ and SSIM↑ on both *UFSC*/*UFUC*, and obtains the best FID↓ scores on the large (*L*) and small (*S*) datasets. With NFA-8 and SE, GAR-Font further improves in both style fidelity and structural accuracy, achieving RMSE↓ ($0.2503/0.2540$) and SSIM↑ ($0.6411/0.6356$) on *UFSC*/*UFUC*, surpassing existing methods by a large margin.

Table 1. Quantitative results on vision-only FFG. ■ / ■ denote the best results on the *Large* / *Small* datasets, respectively. GAR-Font(I_8) shows competitive metrics at pretraining and achieves top reconstruction and perceptual performance on the Large UFSC split after NFA-8+SE, demonstrating improved structural fidelity and perceptual quality.

Method	Train	Unseen Fonts Seen Characters (UFSC)						Unseen Fonts Unseen Characters (UFUC)					
	Set	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑
LF-Font	*S*	0.3984	0.3276	0.2641	32.7464	0.4624	0.0148	0.3983	0.3299	0.2620	32.7028	0.4889	0.0160
	L	0.3988	0.3318	0.2450	21.1028	0.8989	0.0024	0.3986	0.3333	0.2451	21.8141	0.9082	0.0028
VQ-Font	*S*	**0.2727**	**0.5642**	0.1830	35.2472	0.8763	0.0016	**0.2744**	**0.5616**	0.1822	36.7914	0.8882	0.0016
	L	0.2734	0.5633	0.1749	19.3103	0.8549	0.0014	0.2741	0.5627	0.1746	19.971	0.8434	0.0015
DG-Font	*S*	0.3208	0.4991	0.1281	13.8392	0.9706	0.0764	0.3173	0.5074	0.1270	14.8249	0.9706	0.0797
	L	0.3117	0.5193	0.1235	17.1646	0.9172	0.1089	0.3074	0.5289	0.1214	17.2638	0.9174	0.1120
CF-Font	*S*	0.3110	0.526	0.1301	18.6961	0.8542	0.0725	0.3077	0.5333	0.1282	19.309	0.8687	0.0747
	L	0.2993	0.5418	0.1155	13.354	0.8931	0.1549	0.2967	0.5474	0.1144	14.0878	0.8947	0.1570
IF-Font	*S*	0.4076	0.3220	0.1713	14.2393	0.9804	0.0246	0.4063	0.3257	0.1724	14.8211	0.9750	0.0211
	L	0.3969	0.3374	0.1480	11.6470	0.9387	0.1148	0.3949	0.3433	0.1476	11.8445	0.9354	0.1153
Diff-Font	*S*	0.3688	0.3903	0.1851	10.5722	0.4419	0.0515	-	-	-	-	-	-
	L	0.3651	0.3949	0.1791	9.7051	0.4029	0.1208	-	-	-	-	-	-
Font-Diffuser	*S*	0.3010	0.4994	0.1728	26.2647	**0.9994**	0.0212	0.2999	0.5023	0.1720	26.9122	**0.9999**	0.0226
	L	0.2645	0.5813	0.1419	21.4246	**0.9979**	0.0527	0.2631	0.5849	0.1407	21.9637	**0.9980**	0.0578
GAR-Font(I_8)	*S*	0.3080	0.5052	0.1313	7.9484	0.9408	0.0802	0.3142	0.4932	0.1421	8.4841	0.8993	0.0796
	L	0.2772	0.5799	0.1112	7.7155	0.9146	0.1928	0.2784	0.5787	0.1121	8.0349	0.8912	0.1892
GAR-Font	*S*	0.2979	0.5418	0.1177	**6.6909**	0.9195	**0.1128**	0.3002	0.5354	0.1195	**6.4693**	0.9191	**0.1112**
(I_8, +NFA-8)	*L*	0.2600	0.6158	0.0979	**6.5634**	0.9210	0.3313	0.2603	0.6160	0.0983	**6.5842**	0.8921	0.3518
GAR-Font	*S*	0.2909	0.5619	**0.1111**	8.4951	0.9817	0.1101	0.2935	0.5553	**0.1129**	8.3504	0.9804	0.1025
(I_8, +NFA-8+SE)	*L*	**0.2503**	**0.6411**	**0.0885**	8.9851	0.9795	**0.3518**	**0.2540**	**0.6356**	**0.0903**	8.6602	0.9670	**0.3735**



Figure 5. Qualitative results on vision-only FFG (UFSC, *Large* dataset). □ / □ indicate structural errors and style mismatches. GAR-Font(I_8, +NFA-8+SE) produces the most faithful glyphs with superior structure and style alignment.

Figure 5 presents comparisons under the *UFSC* setting. GAR-Font(I_8, +NFA-8+SE) generates the most precise and coherent glyphs. By contrast, LF-Font, VQ-Font, DG-Font, CF-Font, and Diff-Font often exhibit stroke distortion or structural collapse on complex styles. IF-Font frequently produces incomplete glyphs with overly thick strokes. Font-Diffuser maintains reasonable structure and style but often shows slight style shifts in the results.

4.4. Efficient Vision-Language Adaptation

We further evaluate **multimodal FFG** to assess whether textual style descriptions can improve over vision-only references. Using GAR-Font(I_8) on the large (*L*) dataset, we vary the number of visual references $n_{\text{ref}} \in 2, 4, 8$ as baselines. We then introduce multimodal variants GAR-Font(M_2) and GAR-Font(M_4), which pair 2 or 4 visual references with one textual style description. For experimental evaluation, due to the lack of such corpora, Qwen2.5-VL [3] generates descriptions as a proxy of human-authored design intent; details are provided in the supplementary material.

As shown in Table 2, both multimodal models outperform their vision-only counterparts with the same number of visual references across all major metrics. GAR-Font(M_4) even surpasses the 8-reference visual model, achieving lower RMSE↓/LPIPS↓ and higher SSIM↑ on the *UFSC*, along with a better FID↓ (7.4915 vs. 7.7155). Similar gains appear on *UFUC*, confirming that language provides complementary style cues and reduces reliance on numerous visual references. A mild decrease in ACC(S) is observed, likely because textual guidance yields smoother and

Table 2. Quantitative results on multimodal FFG. GAR-Font(M_2) and GAR-Font(M_4) integrate textual style guidance with 2 or 4 visual references, outperforming vision-only baselines across all major metrics. Results demonstrate that language complements visual references, enhancing fine-grained style representation while reducing reliance on handcrafted visuals.

Method	Unseen Fonts Seen Characters (UFSC)						Unseen Fonts Unseen Characters (UFUC)					
	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑
$n_{\text{ref}} = 2$	0.2816	0.5695	0.1158	7.3553	0.9184	0.1535	0.2825	0.5694	0.1162	7.5500	0.8930	0.1619
$n_{\text{ref}} = 4$	0.2807	0.5735	0.1138	7.3781	0.9206	0.1741	0.2813	0.5735	0.1146	**7.4880**	0.8958	0.1818
$n_{\text{ref}} = 8$	0.2772	0.5799	0.1112	7.7155	0.9146	**0.1928**	0.2784	0.5787	0.1121	8.0349	0.8912	**0.1892**
GAR-Font(M_2)	0.2811	0.5724	0.1136	**7.3145**	**0.9296**	0.1203	0.2817	0.5731	0.1143	7.5306	**0.9068**	0.1289
GAR-Font(M_4)	**0.2764**	**0.5825**	**0.1098**	7.4915	0.9260	0.1688	**0.2776**	**0.5816**	**0.1107**	7.6607	0.9039	0.1744



Figure 6. Qualitative results on multimodal FFG(UFSC, *Large* dataset). ▢ denotes local slight structural mistakes and ▢ marks samples with stylistic regression. With textual style guidance, GAR-Font(M_2) and GAR-Font(M_4) yield more structurally accurate and stylistically coherent generations, demonstrating enhanced stable fine-grained style control compared to vision-only baselines (*first three rows*).

more diverse styles beyond the classifier's limit.

Qualitative examples in Figure 6 show that GAR-Font(M_2) and GAR-Font(M_4) preserve styles more reliably than vision-only models, particularly when $n_{\text{ref}} = 2$, where the visual baseline noticeably drifts toward generic shapes. This highlights the effectiveness of language in reinforcing style fidelity under low-reference conditions.

4.5. Ablation Studies

4.5.1. On G-Tok's Hybrid Architecture

To validate G-Tok's hybrid CNN-ViT design, we start from a CNN-based tokenizer [52] and progressively insert ViT blocks at different depths to introduce global awareness.

We conduct two complementary analyses on *UFUC* test:

1. **Linear Probing** evaluates the discriminative quality of frozen G-Tok representations for style and content prediction using a single-layer linear classifier. Features are extracted and flattened from the frozen tokenizer encoder, and high classification accuracy indicates that the encoder effectively captures and discriminates structural and stylistic information.
2. **Reconstruction Robustness** recovers glyphs corrupted by localized Gaussian noise ($\sigma = 0.2$, affecting 20% of the glyph area). High performance indicates that the tokenizer effectively preserves global structure and stylistic cues despite local perturbations.

Table 3 shows that the pure ViT excels in linear probing but suffers in reconstruction, while the CNN baseline ensures stable recreation with weaker probing results. The hybrid G-Tok progressively integrates global attention, with the CNN-ViT-6 variant achieving the best, improving both classification accuracy and reconstruction fidelity across all metrics. Visualizations are in the supplementary material.

Table 3. Ablation of G-Tok's hybrid CNN-ViT architecture on UFUC(*Small* dataset). **Bold** and underline indicate the best and second-best results. Progressive integration of attention improves both discriminative representation and reconstruction fidelity.

Method	Linear Probing		Reconstruction Robustness			
	Acc(S)↑	Acc(C)↑	RMSE↓	SSIM↑	LPIPS↓	FID↓
CNN	0.5515	0.3879	0.1167	0.8535	0.0423	28.4279
ViT-6	**0.6907**	**0.5334**	0.1636	0.7333	0.0933	98.4270
CNN-ViT-2	0.5229	0.3772	0.1114	0.8552	0.0420	30.0034
CNN-ViT-4	0.5585	0.3667	0.1114	0.8563	0.0430	24.7323
CNN-ViT-6	0.6277	0.4897	**0.1088**	**0.8594**	**0.0412**	**22.1577**

Table 4. Ablation of G-Tok's architecture on UFUC(*Small* dataset). Incorporating self-attention (CNN+Non-Causal ViT) improves global modeling while causal attention further strengthens sequential modeling, yielding the best overall performance.

Method	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑
CNN	0.3447	0.4350	0.1728	10.5239	0.6722	0.0221
CNN+Non-Causal ViT	0.3271	0.4745	0.1562	8.7504	0.8019	0.0436
CNN+Causal ViT	**0.3142**	**0.4932**	**0.1421**	**8.4841**	**0.8993**	**0.0796**

Table 5. Ablation of decoding strategy and pixel-level supervision on UFSC and UFUC(*Small* dataset). Soft decoding outperforms hard decoding across all metrics, and pixel supervision further enhances reconstruction fidelity and recognition accuracy.

Training Loss	Decoding	Unseen Fonts Seen Characters (UFSC)						Unseen Fonts Unseen Characters (UFUC)					
	Strategy	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑
w/o pixel loss	hard	0.3235	0.4679	0.1517	10.3181	0.8647	0.0377	0.3322	0.4510	0.1602	11.2142	0.7465	0.0396
	soft	0.3231	0.4745	0.1502	9.6696	0.9171	0.0426	0.3313	0.4583	0.1589	10.3813	0.8104	0.0430
w/ pixel loss	hard	0.3083	0.4991	0.1329	8.3024	0.9032	0.0771	0.3157	0.4862	0.1448	8.9043	0.8404	0.0680
	soft	**0.3080**	**0.5052**	**0.1313**	**7.9484**	**0.9408**	**0.0802**	**0.3142**	**0.4932**	**0.1421**	**8.4841**	**0.8993**	**0.0796**

Table 6. Quantitative comparison of joint training (GAR-Font(VL_k)) versus the decoupled adapter scheme (GAR-Font(M_k)) on unseen fonts. The decoupled Language-style Adapter surpasses joint-trained multimodal encoders, validating more effective vision–language alignment and improved reconstruction and perceptual metrics.

Method	Unseen Fonts Seen Characters (UFSC)						Unseen Fonts Unseen Characters (UFUC)					
	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑	RMSE↓	SSIM↑	LPIPS↓	FID↓	Acc(C)↑	Acc(S)↑
GAR-Font(VL_2)	0.3070	0.5124	0.1458	10.7566	0.8552	0.1104	0.3100	0.5087	0.1473	11.0381	0.7979	0.1209
GAR-Font(VL_4)	0.2983	0.5378	0.1279	7.9272	0.8970	0.1165	0.3006	0.5351	0.1290	8.1126	0.8665	0.1216
GAR-Font(M_2)	0.2811	0.5724	0.1136	**7.3145**	**0.9296**	0.1203	0.2817	0.5731	0.1143	**7.5306**	**0.9068**	0.1289
GAR-Font(M_4)	**0.2764**	**0.5825**	**0.1098**	7.4915	0.9260	**0.1688**	**0.2776**	**0.5816**	**0.1107**	7.6607	0.9039	**0.1744**

4.5.2. On G-Tok's Global and Causal Modeling

We further ablate G-Tok's ViT architecture to assess its global and causal modeling. We pretrain three visual AR variants that differ in G-Tok and examine them on *UFUC*:

1. **CNN:** a baseline CNN tokenizer without global context;
2. **CNN + Non-causal ViT:** a hybrid CNN-ViT tokenizer with self-attention that only supports global interaction;
3. **CNN + Causal ViT:** full G-Tok, hybrid CNN-ViT tokenizer with causal self-attention for sequential modeling.

Table 4 shows that incorporating self-attention ViT improves performance over the CNN baseline, while the causal ViT further enhances sequential modeling, achieving the best results across all metrics. Detailed *UFSC* results are provided in the supplementary material.

4.5.3. On AR Generator's Soft-decoding

To validate the effectiveness of pixel-level supervision and the soft decoding strategy, we conduct ablation experiments on our vision-only generator. The results in Table 5 demonstrate that soft decoding achieves superior performance over all metrics, particularly when combined with pixel-level supervision. This combination yields the lowest RMSE↓, LPIPS↓, and FID↓ while achieving the highest SSIM↑ and classifier accuracies on *UFSC* and *UFUC*, indicating improved pixel-level fidelity and faithful style preservation.

4.5.4. On Multimodal Style Encoder's Adaptation

GAR-Font adopts a decoupled training paradigm: a visual encoder is pretrained, followed by multimodal control via a lightweight adapter. To validate this design, we compare it with jointly training a multimodal style encoder under the same configuration on the large (*L*) dataset.

We evaluate two joint training variants, **GAR-Font(VL_2)** and **GAR-Font(VL_4)**, against the decoupled multimodal variants **GAR-Font(M_2)** and **GAR-Font(M_4)**. As Table 6 shows, decoupled variants consistently outperform joint-trained counterparts under the same number of visual references. For example, GAR-Font(M_4) attains the best RMSE↓, LPIPS↓, and SSIM↑ on *UFSC* / *UFUC*, while GAR-Font(VL_4) performs worse. GAR-Font(M_2) also achieves the best FID↓ on *UFSC* (7.3145). These results validate the decoupled vision–language training paradigm, where a pretrained visual encoder and a lightweight adapter achieve efficient and robust text–visual alignment, improving generalization in multimodal FFG.

4.6. Limitations and Future Work

Although GAR-Font surpasses existing FFG methods in visual fidelity, several limitations remain. First, the Multimodal Style Encoder currently relies on late adaptation via a language-style adapter; exploring earlier text-image fusion may enable finer stylistic control and reduce reliance on visual references. Second, our experiments are conducted at a resolution of 64×64, which may limit direct applicability to high-DPI scenarios; extending the framework to higher resolutions may require G-Tok to handle longer token sequences. Finally, we aim to expand controllability beyond style and content, adding attributes like stroke thickness, character width, and slant for more flexible font generation.

5. Conclusion

In this paper, we present GAR-Font, a global-aware autoregressive framework for few-shot font generation that combines a hybrid global tokenizer, an autoregressive generator with a multimodal style encoder, and a post-refinement pipeline. GAR-Font achieves superior structural and stylistic fidelity, outperforming prior vision-only baselines, while demonstrating that language-guided adaptation can rival or exceed heavy visual conditioning with improved flexibility.

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